

A Comprehensive Benchmark of Deep Learning Architectures for 2D Echocardiographic Segmentation: From CNNs to Transformers

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Abstract

Two-dimensional echocardiography is the most widely used cardiac imaging modality, but quantitative analysis is hampered by speckle noise, low boundary contrast, and inter-observer variability. U-Net descendants, attention-augmented CNNs, hybrid CNN–Transformer designs, and pure Transformer networks all report strong results on this task, yet direct comparison between them is confounded by inconsistent training protocols and evaluation pipelines. We benchmark nine representative architectures spanning three paradigms under one training protocol, one data split, and one evaluation pipeline on the CAMUS dataset, reporting per-class Dice and IoU, boundary distances in millimetres, biplane Simpson’s ejection fraction (EF) error, phase-stratified and image-quality-stratified results, efficiency, and Bonferroni-corrected Wilcoxon tests. Two findings stand out. First, boundary accuracy has saturated: the best architecture reaches HD95 = 4.08 mm against a published CAMUS baseline of ≈ 4.3 mm, and the three best architectures lie within 0.2 mm of one another, well inside training variance. Clinical agreement is bounded by the measurement rather than by the network: applying the official CAMUS biplane pipeline to the ground-truth masks gives EF MAE = 5.44%, and the best architecture reaches 6.62%. Segmentation therefore contributes about 1.2 points of EF error on top of a 5.4-point floor that no model can cross, and the spread across all nine architectures is 4.0 points. Neither axis leaves much headroom for architectural change. Second, we identify a reporting error that makes cross-paper boundary comparison unreliable. Computing Hausdorff distances on the network’s resized grid while applying the native image spacing under-reports every distance by the resampling factor — on CAMUS, $\approx 2.2\times$. Under that convention our own models score below 2 mm, which would appear to halve the published baseline; scored at native resolution the same predictions merely match it. Any sub-2 mm HD95 reported on CAMUS warrants checking

against this failure mode. Code, checkpoints, and per-patient predictions are publicly released.

Keywords: medical image segmentation, echocardiography, CAMUS, deep learning benchmark, U-Net, Swin Transformer, TransUNet

1. Introduction

Two-dimensional echocardiography is the most widely used cardiac imaging modality worldwide: it is portable, inexpensive, non-ionising, and provides real-time visualisation of cardiac structure and function. Its core quantitative output — left ventricular ejection fraction (LVEF), computed from end-diastolic and end-systolic volumes via the biplane Simpson’s method [18] — drives a large fraction of clinical cardiac decisions, from diagnosis of heart failure to monitoring of cardiotoxic chemotherapy. The clinical importance of LVEF is matched by the operator-dependent nature of its measurement: published inter-observer LVEF variability is approximately 7–10% [19], which directly limits the sensitivity of LVEF as a clinical decision boundary.

Automated segmentation of the left ventricular endocardium (LV-endo), epicardium (LV-epi), and left atrium (LA) from two-dimensional apical 2-chamber (2CH) and 4-chamber (4CH) views would, in principle, eliminate this operator-dependent variability and provide reproducible volumetric measurements. The technical challenge is substantial: ultrasound suffers from low signal-to-noise ratio, speckle noise, low boundary contrast, acoustic shadowing, and view-dependent geometric foreshortening.

Architectural pluralism.. Three paradigms have emerged as competitive solutions in this space:

- **Pure convolutional networks.** U-Net [35] and its descendants (residual encoders, SE blocks, attention gates, deep supervision) [28, 29] remain the dominant architectural family, with nnU-Net [11] serving as a strong auto-configured baseline.
- **Hybrid CNN–Transformer networks.** TransUNet [3] and similar hybrids combine a convolutional encoder with a Vision Transformer bottleneck, explicitly trading off local detail (CNN) against global context (self-attention).

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- **Pure Transformer networks.** Swin-UNet [2] replaces the convolutional *backbone* with a hierarchy of windowed self-attention blocks — retaining only a shallow convolutional patch-embedding stem — demonstrating that predominantly attention-based encoders can match or exceed CNN performance on a range of medical tasks.

The benchmarking gap. Despite this architectural pluralism, the literature offers no systematic, like-for-like comparison of these three paradigms on a single shared evaluation pipeline. Each paper introducing a new architecture typically compares against a small number of baselines under different training protocols, augmentations, learning-rate schedules, and evaluation metrics. As a result it is difficult for clinicians or practitioners to answer a basic question: *which architectural family should I use for CAMUS-style apical 2D echocardiographic segmentation, and what are the trade-offs?*

This paper provides that comparison on the CAMUS dataset [19], the standard public 2D echocardiographic segmentation benchmark, with 500 patients across two views and two cardiac phases. We train nine representative architectures spanning the three paradigms above using one identical protocol — same optimiser, loss, schedule, augmentation, mixed-precision settings, and early-stopping criterion — and evaluate them on a single shared pipeline reporting geometric, boundary, clinical, and efficiency metrics simultaneously.

Contributions.

- We present the first unified, like-for-like benchmark of deep segmentation architectures on CAMUS across the three major paradigms: nine architectures spanning pure CNN, hybrid CNN–Transformer, and pure Transformer, trained and evaluated under a single shared protocol.
- We report *multi-dimensional* evaluation, going beyond Dice to include HD95 in millimetres (with per-sample NIfTI pixel spacing), ASSD, biplane Simpson’s EF MAE and Pearson r , ED/ES stratified metrics, robustness across CAMUS image-quality grades, and efficiency benchmarks (parameters, FLOPs, latency, peak memory). The full results matrix and per-patient outputs are released.
- We apply pairwise Wilcoxon signed-rank tests with Bonferroni correction to identify which architectural differences are statistically significant, and produce a critical-difference diagram showing the top-tier band of equivalent models.

- **We document a boundary-metric reporting error that makes published HD95 values on CAMUS mutually incomparable.** Computing the distance transform on the network’s resized 256×256 grid while attaching the native NIfTI spacing under-reports every distance by the resampling factor, which on CAMUS is $\approx 2.2\times$. The effect is large enough to invert a conclusion: under that convention our models score 1.8–2.9 mm and appear to halve the published baseline, while the identical predictions scored at native resolution give 4.1–5.7 mm and merely match it. We report both, and recommend that boundary distances be computed at acquisition resolution — a concrete instance of the validation concerns raised by Reinke et al. [33] and Maier-Hein et al. [26].
- Our best architecture, TransUNet, reaches Dice = 0.9121, HD95 = 4.08 mm; nnU-Net reaches Dice = 0.9099, HD95 = 4.27 mm with $6.5\times$ fewer parameters and roughly $3\times$ faster inference. Boundary accuracy under our unified protocol reproduces the originally published nnU-Net baseline (≈ 4.3 mm) [19] rather than improving on it, and EF is floor-limited in the same way: the ground-truth oracle scores 5.44% MAE under an identical pipeline against the best model’s 6.62%, leaving roughly 1.2 points attributable to segmentation at all.
- We release a fully reproducible benchmark framework: training, evaluation, statistical analysis, and figure generation code.

Outline. Section 2 reviews segmentation architectures and the CAMUS benchmark. Section 3 describes the nine architectures, the unified training protocol, and the evaluation pipeline. Section 4 reports the experimental results. Section 5 discusses architectural design choices, trade-offs, and limitations. Section 6 concludes.

2. Related Work

2.1. Convolutional segmentation networks

U-Net [35] established the encoder–decoder architecture with skip connections that remains the dominant template for medical image segmentation. Subsequent work has refined the encoder (residual blocks / ResNet [8] and later compound-scaled backbones), the decoder (transposed convolution vs. bilinear upsampling, dense connections [10]), and the skip mechanism (attention gates [29], dense skips). Squeeze-and-excitation blocks [9] provide channel-wise recalibration. Atrous spatial pyramid pooling (ASPP) in DeepLabV3+ [4] replaces the deep en-

coder with dilated convolutions to enlarge the receptive field without down-sampling. Feature Pyramid Networks [21] aggregate features across multiple scales for dense prediction.

The nnU-Net framework [11] integrates these architectural choices with automatic data-dependent hyperparameter selection (target spacing, patch size, batch size, augmentation pipeline) and has set strong baselines on numerous medical benchmarks including CAMUS.

2.2. Transformer-based segmentation networks

Self-attention’s quadratic complexity made global attention impractical for high-resolution images until Vision Transformers [6] demonstrated patch-based tokenisation. TransUNet [3] was the first work to bring this design to medical segmentation, combining a CNN encoder (typically ResNetV2-BiT [17]) with a ViT-B/16 bottleneck and a cascaded upsampler decoder. The hybrid design retains the inductive bias of convolutions at high resolution while granting global context at the bottleneck.

Swin Transformer [23] introduced windowed self-attention with shifted windows, restoring linear complexity in image size. Swin-UNet [2] adapts this to segmentation with a fully Transformer-based U-Net. SwinV2 [22] and Swin-MAE [42] further refine pretraining strategies.

2.3. Echocardiographic segmentation and CAMUS

The CAMUS dataset [19] (*Cardiac Acquisitions for Multi-structure Ultrasound Segmentation*) provides 2D apical-2-chamber and apical-4-chamber acquisitions from 500 patients, each annotated by an expert echocardiographer at end-diastole and end-systole with four classes (background, LV-endo, LV-epi, LA). The official split assigns 400 patients to training, 50 to validation, and 50 to test, with test annotations held out by the challenge organisers. Image quality is graded into *Good*, *Medium*, and *Poor* categories by the same experts.

The original CAMUS paper reported nnU-Net achieving $HD \approx 4.3$ mm on LV-endo and $\sim 9\%$ EF MAE. Subsequent work [31, 36] has reported incremental improvements using deeper architectures, custom losses, and view-aware training. Several papers have additionally explored sequence-aware models on CAMUS half-cycle annotations [31], but these are out of scope for the present benchmark.

Activity on echocardiographic segmentation has continued along four largely separate lines. The first pursues *architectural* refinement: bi-directional token matching between frames [24],

dynamic-learning U-Nets that enforce temporal consistency across the cardiac cycle [32], contour-guided query-based fusion aimed explicitly at boundary quality [39], and selective state space models applied to paediatric echocardiography [43]. The second addresses *domain shift*, which is the dominant obstacle to deploying any of these models across scanners and sites: reinforcement-learning-based domain adaptation [14, 13] and interpretable quantification of the shift itself [7]. The third applies *foundation models* and self-supervision, either by adapting general-purpose promptable segmenters to medical images [25], by generating synthetic cardiac ultrasound for augmentation [38, 34], or by asking what self-supervised echocardiographic representations actually transfer [27]. The fourth targets the *clinical endpoint* directly, predicting or explaining ejection fraction rather than the mask that produces it [44, 37, 20, 41].

What these lines share is a comparison protocol inherited from whichever baseline was convenient. Each reports its own training recipe, its own resolution, and — as we show in Section 4.2 — not always the same definition of a millimetre. That is the gap this benchmark addresses.

EchoNet-Dynamic [30] provides a complementary large-scale video dataset for EF regression. We focus on CAMUS as the standard benchmark for *segmentation-derived* EF. Because EchoNet-Dynamic annotates only the left-ventricular endocardium in a single apical view, it cannot exercise the LV-epicardium, left-atrium, or biplane-EF components of our protocol; whether the architectural trade-offs reported here transfer to other echocardiographic cohorts is therefore a question we leave to future cross-dataset work rather than a claim we make here.

2.4. Validation practice and metric reporting

Running alongside the architectural literature is a smaller body of work arguing that the field’s *measurement* practice, not its model design, is now the limiting factor. Reinke et al. [33] catalogue the ways standard segmentation metrics mislead when applied without regard to structure size, shape, or resolution, and the Metrics Reloaded framework [26] turns that catalogue into problem-driven recommendations for choosing and reporting them. Isensee et al. [12] show that much of the reported progress in 3D medical segmentation dissolves under rigorous validation, with a well-configured U-Net still matching or beating architectures claimed to supersede it. Kazaj et al. [15] reach a similar conclusion for the CNN-versus-Transformer-versus-Mamba question specifically, and Urooj et al. [40] audit a segmentation subfield and find that reported rankings shift once evaluation inconsistencies are removed.

Boundary metrics are especially exposed. Hausdorff distances are reported in millimetres, but a millimetre is only defined once the grid on which the distance transform is computed and the physical spacing attached to that grid refer to the same image. Networks are trained at a fixed resolution while clinical images are not, so the two can silently diverge. We quantify the consequence on CAMUS in Section 4.2: the same predictions score 1.8 mm or 4.1 mm depending on which convention is used, and only the latter is comparable to the published baseline.

2.5. What is missing

A small number of recent works compare a handful of architectures on CAMUS in their experimental sections, but none provide:

- A unified training protocol across CNN, hybrid, and Transformer paradigms (papers typically inherit each architecture’s own published hyperparameters, confounding architecture with protocol).
- Per-class HD95 in millimetres with per-sample NIfTI spacing (most papers report Dice only, or use pixel-space HD95).
- ED- versus ES-stratified reporting (essential for biplane Simpson’s EF analysis).
- Robustness analysis by CAMUS image-quality grade.
- Pairwise Wilcoxon significance with Bonferroni correction and a critical-difference diagram.
- Released model checkpoints and per-patient predictions for independent reanalysis.

This paper aims to fill each of these gaps.

3. Materials and Methods

3.1. Dataset and splits

We use the CAMUS dataset [19], comprising 500 patients imaged in apical-2-chamber (2CH) and apical-4-chamber (4CH) views, with expert annotations at end-diastole (ED) and end-systole (ES). Each frame is labelled with four classes: background, LV-endocardium, LV-epicardium, and left atrium. Per-patient pixel spacing is recorded in the NIfTI header and ranges between approximately 0.30 and 0.42 mm.

We adopt the official CAMUS split: 400 patients for training, 50 for validation, and 50 for testing, with no overlap. After ED+ES frame extraction this yields approximately 1,600 training, 200 validation, and 200 test images. Half-cycle sequences are optionally used for augmentation during training; results are reported on the canonical ED+ES test set only.

3.2. Architectures

We benchmark nine representative architectures spanning three paradigms:

Pure CNN (seven architectures).

- **UNet-V1** [35]: canonical encoder–decoder with skip connections and trainable transposed-convolution upsampling.
- **UNet-V2** (custom): residual blocks, squeeze-and-excitation [9], attention-gated skips [29], and optional deep supervision.
- **UNet-ResNet**: U-Net with a ResNet-34 encoder [8] pretrained on ImageNet.
- **DeepLabV3+** [4]: ResNet-50 encoder, atrous spatial pyramid pooling bottleneck, lightweight decoder.
- **nnU-Net** [11]: auto-configured U-Net variant, trained from scratch.
- **DenseContextU-Net** (custom): dense blocks [10] with a global-context module in the bottleneck.
- **FPN-UNet**: U-Net decoder with a Feature Pyramid [21] neck over a ResNet-50 encoder.

Hybrid CNN–Transformer (one architecture). **TransUNet** [3]: ResNetV2-BiT encoder [17] feeding a ViT-B/16 bottleneck, with a cascaded upsampler decoder using skip connections from encoder stages.

Pure Transformer (one architecture). **Swin-UNet** [2]: Swin-Tiny encoder [23] pretrained on ImageNet-1k, symmetric Transformer decoder, patch-expanding upsampling.

A complete summary of architectural choices is given in Table 1. We deliberately exclude all SSM-based variants (Mamba, Mamba-2, VMamba) from the present benchmark; those are the subject of a companion study.

3.3. Training protocol

All nine architectures share a single training protocol:

- AdamW optimiser ($\beta_1=0.9$, $\beta_2=0.999$, weight decay 10^{-4}), initial learning rate 10^{-4} , cosine annealing with warm restarts ($T_0=50$ epochs).
- Composite loss: $\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}} + 0.1 \mathcal{L}_{\text{boundary}}$, where the boundary term is computed on the signed distance transform [16].

Table 1: The nine architectures benchmarked in this paper. “Pretrained” indicates whether the encoder is initialised from ImageNet/BiT pretrained weights or trained from scratch.

Paradigm	Architecture	Encoder	Decoder	Pretrained	Params (M)
Pure CNN	UNet-V1	scratch / conv	transposed conv	—	31.04
Pure CNN	UNet-V2	scratch / conv + SE	attn-gated skips	—	33.00
Pure CNN	UNet-ResNet	ResNet-34	U-Net decoder	ImageNet-1k	29.07
Pure CNN	DeepLabV3+	ResNet-50	ASPP + light dec.	ImageNet-1k	40.34
Pure CNN	nnU-Net	auto-configured	deep-supervision	—	15.62
Pure CNN	DenseContextU-Net	dense + context	dense decoder	—	2.18
Pure CNN	FPN-UNet	ResNet-50	FPN + U-Net dec.	ImageNet-1k	32.16
Hybrid	TransUNet	ResNetV2 + ViT-B/16	cascaded upsampler	BiT + IN-21k	102.09
Pure Transformer	Swin-UNet	Swin-Tiny	Swin decoder	IN-1k	41.84

- Image size 256×256 for all CNN and hybrid models; 224×224 for Swin-UNet, which requires input divisible by $\text{window_size} \times 2^{n_{\text{stages}}} = 7 \times 32 = 224$.
- Augmentation: horizontal flip, ± 10 deg rotation, small elastic deformation, mild brightness/contrast jitter.
- Up to 100 epochs, automatic mixed precision (AMP, FP16), gradient clipping at $\|\nabla\|_2 \leq 1$ for pretrained-encoder models (Swin-UNet, TransUNet) to prevent the early-epoch divergence we otherwise observed.
- Early stopping with patience 20 on validation mean Dice.
- Training hardware: Google Colab Pro NVIDIA L4 (24 GB) and A100 (40 GB) instances. The final test-set evaluation reported in Section efsec:results was run in a single pass on one NVIDIA RTX 4060 Laptop GPU, so that every metric in every table comes from the same inference run.

All nine architectures were trained with a *uniform* batch size of 32 — the largest value that fits every model within the training-GPU memory budget — rather than a per-architecture batch size, so that the optimisation regime is held as constant as possible across the comparison. We did *not* perform per-architecture hyperparameter search: the goal is explicit cross-architecture comparison under one shared protocol.

3.4. Evaluation metrics

We report a multi-dimensional set of metrics:

Geometric.. Per-class Dice and Intersection-over-Union (IoU) for LV-endo, LV-epi, and LA, computed on the full test set and separately for ED and ES frames.

Boundary.. 95th-percentile Hausdorff distance (HD95) and average symmetric surface distance (ASSD), both in *millimetres* using the per-sample pixel spacing recorded in NIfTI headers. This is consistent with the official CAMUS challenge protocol and is essential for cross-paper comparison.

Clinical.. Biplane Simpson’s ejection fraction (EF) computed from the LV-endo masks at ED and ES across paired 2CH and 4CH views. We report EF MAE (%), Pearson correlation r with the ground-truth EF, and Bland–Altman bias and 95% limits of agreement [1].

Phase-stratified.. All metrics are additionally reported separately for ED and ES frames, which is informative because ED frames tend to be larger and have clearer endocardial boundaries.

Quality-stratified robustness.. Each CAMUS patient carries an expert-assigned image-quality grade (Good / Medium / Poor). A quality-stratified robustness breakdown is deferred to the extended version of this work.

Statistical significance.. For every pair of models we run a Wilcoxon signed-rank test on per-sample mean Dice across the test set, with Bonferroni correction across the $\binom{9}{2} = 36$ pairs. We additionally produce a critical-difference diagram [5] illustrating the band of statistically equivalent top models.

Efficiency.. Trainable parameters, GFLOPs at 256×256 input (measured with `fvcore` on CUDA), single-image inference latency (mean over 100 forward passes after 10 warmup, batch size 1, FP16), and peak GPU memory during forward+backward.

3.5. Reproducibility

A single shared evaluation pipeline is used for all models; the same code path generates the JSON outputs, CSV tables, LaTeX tables, and the figures in this paper. The training pipeline

Table 2: Mean and per-class Dice on the CAMUS official test split for the nine base architectures. The bracketed interval is the bootstrap 95% confidence interval on mean Dice over the 50-patient test set. Best per column in **bold**, second-best underlined. Per-class scores are LV-endocardium, LV-epicardium, and left atrium.

Architecture	Paradigm	Mean Dice	95% CI	LV-endo	LV-epi	LA	Params (M)
TransUNet	Hybrid	0.9121	[0.9078, 0.9162]	0.9367	0.8789	0.9208	102.1
nnU-Net	CNN	<u>0.9099</u>	[0.9054, 0.9143]	0.9371	0.8790	0.9136	15.6
UNet-V1	CNN	0.9085	[0.9041, 0.9130]	0.9369	0.8757	0.9129	31.0
UNet-ResNet	CNN	0.9081	[0.9019, 0.9136]	0.9341	0.8742	0.9160	29.1
UNet-V2	CNN	0.8963	[0.8913, 0.9015]	0.9239	0.8592	0.9059	33.0
Swin-UNet	Transformer	0.8882	[0.8829, 0.8936]	0.9172	0.8520	0.8954	41.8
DeepLabV3+	CNN	0.8602	[0.8511, 0.8681]	0.9109	0.8187	0.8510	40.3
FPN-UNet	CNN	0.8596	[0.8526, 0.8661]	0.8962	0.7944	0.8881	32.2
DenseContextU-Net	CNN	0.8505	[0.8414, 0.8604]	0.8977	0.8046	0.8492	2.2

fixes the random seed (42) across PyTorch, NumPy, and Python. Per-model checkpoints, per-test-patient prediction NIfTI files, and per-sample Dice arrays are released to allow independent re-analysis without re-running training.

4. Results

We organise the results into six subsections. Section 4.1 gives the main test-set leaderboard. Sections 4.2 and 4.3 report boundary and clinical (EF) metrics. Section 4.4 examines robustness by image quality. Section 4.5 reports the accuracy $\times$ efficiency trade-off. Section 4.6 reports statistical significance.

4.1. Main test-set Dice

Table 2 summarises mean Dice, per-class Dice, and the parameter count for each architecture on the official CAMUS test split.

Headline ranking. **TransUNet** achieves the highest mean Dice (0.9121), narrowly ahead of **nnU-Net** (0.9099, $\Delta = 0.0022$), **UNet-V1** (0.9085), and **UNet-ResNet** (0.9081). The four top architectures are tightly clustered within $\Delta\text{Dice} \leq 0.004$ — within the Bonferroni-corrected statistical-significance band (Section 4.6), so their ordering should not be read as a ranking.

Table 3: Intersection-over-union on the CAMUS test split, mean and per class. IoU orders the architectures almost identically to Dice, as expected of two monotonically related overlap measures; it is reported because IoU is the more common metric outside medical imaging and makes this benchmark comparable to that literature. Best per column in **bold**.

Architecture	Mean IoU	LV-endo	LV-epi	LA
TransUNet	0.8420	0.8828	0.7864	0.8570
nnU-Net	0.8384	0.8833	0.7864	0.8456
UNet-ResNet	0.8364	0.8790	0.7810	0.8493
UNet-V1	0.8362	0.8828	0.7818	0.8441
UNet-V2	0.8168	0.8610	0.7560	0.8335
Swin-UNet	0.8037	0.8496	0.7447	0.8169
DeepLabV3+	0.7657	0.8389	0.6980	0.7601
FPN-UNet	0.7619	0.8142	0.6631	0.8085
DenseContextU-Net	0.7513	0.8192	0.6783	0.7564

Below this band, **UNet-V2** (0.8963) and **Swin-UNet** (0.8882) form a mid-tier, and **DeepLabV3+** (0.8602), **FPN-UNet** (0.8596), and **DenseContextU-Net** (0.8505) trail.

Per-class.. LV-endo Dice ranges 0.8962 (FPN-UNet) to 0.9371 (nnU-Net); LA Dice ranges 0.8492 (DenseContextU-Net) to 0.9208 (TransUNet); LV-epi is consistently the hardest class for every architecture (mean across models: Dice $\approx$ 0.85, versus $\approx$ 0.92 for LV-endo).

Table 3 reports the same decomposition under intersection-over-union. IoU orders the architectures almost identically to Dice, which is expected of two monotonically related overlap measures; we report it because it is the more common metric outside medical imaging and makes this benchmark comparable to that literature.

4.2. Boundary accuracy (HD95 and ASSD in mm)

Boundary metrics are reported in millimetres using the per-sample NIfTI pixel spacing.

HD95.. TransUNet again leads with HD95 = 4.08 mm, followed by UNet-ResNet (4.26 mm) and nnU-Net (4.27 mm), UNet-V1 (4.62 mm), Swin-UNet (5.08 mm), and UNet-V2 (5.29 mm). All boundary distances are computed at the native image resolution: CAMUS frames are approximately 630×519 with a typical spacing of 0.308 mm, so each prediction is resampled back to

Table 4: Boundary metrics on the CAMUS test split, mean and per class. HD95 is the 95th-percentile Hausdorff distance, ASSD the average symmetric surface distance, both in millimetres computed at native resolution using per-sample NIfTI pixel spacing. The per-class columns show that overlap and distance disagree about which structure is hardest: the epicardium has the lowest Dice for all nine architectures (Table 2), yet the largest HD95 for only two of them — for five the worst boundary is the left atrium. A per-class overlap summary and a per-class distance summary are therefore not interchangeable. Best per column in **bold**.

Architecture	HD95 (mm)				ASSD (mm)			
	Mean	LV-endo	LV-epi	LA	Mean	LV-endo	LV-epi	LA
TransUNet	4.08	3.95	4.20	4.09	0.82	0.80	0.84	0.82
UNet-ResNet	4.26	4.03	4.39	4.37	0.84	0.81	0.87	0.86
nnU-Net	4.27	4.01	4.29	4.51	0.84	0.78	0.84	0.89
UNet-V1	4.62	4.36	4.61	4.89	0.87	0.82	0.88	0.91
Swin-UNet	5.08	4.90	5.00	5.34	1.00	0.98	0.98	1.03
UNet-V2	5.29	5.35	5.29	5.23	1.03	1.04	1.05	1.00
DeepLabV3+	5.71	5.02	5.60	6.51	1.19	1.03	1.17	1.37
FPN-UNet	6.21	6.76	6.66	5.22	1.24	1.26	1.34	1.11
DenseContextU-Net	8.29	7.15	8.21	9.52	1.55	1.37	1.47	1.81

its native grid before the distance transform is taken. Measuring on the 256×256 network grid while applying the native spacing would under-report every distance by the resampling factor of ~ 2.2 , and we caution that this is an easy error to make silently. The three best architectures sit within 0.2 mm of one another (TransUNet 4.08, UNet-ResNet 4.26, nnU-Net 4.27), and the top four within 0.54 mm — a spread well inside the training-variance floor discussed in Section 5. All of them cluster around the published CAMUS nnU-Net baseline of ≈ 4.3 mm [19]. Under a unified modern protocol, then, boundary accuracy on CAMUS appears to have saturated: our reproduction matches the 2019 baseline rather than improving on it. That baseline was in any case reported under the original CAMUS evaluation convention, which differs from our per-sample HD95, so the cross-paper comparison is indicative rather than strictly like-for-like.

ED versus ES. Table 5 reports ED- and ES-stratified mean Dice and HD95. The two phases are closely matched for every architecture (mean Dice within ≈ 0.007 and HD95 within ≈ 1.4 mm), with no consistent advantage to either phase. This is a reassuring property for biplane EF, which

Table 5: End-diastole (ED) and end-systole (ES) stratified mean Dice and HD95 (mm) on the CAMUS test split.

Architecture	Dice		HD95 (mm)	
	ED	ES	ED	ES
nnU-Net	0.9105	0.9093	4.19	4.35
TransUNet	0.9105	0.9137	4.07	4.09
UNet-V1	0.9072	0.9098	4.67	4.57
UNet-ResNet	0.9062	0.9100	4.16	4.37
UNet-V2	0.8989	0.8938	5.06	5.52
Swin-UNet	0.8884	0.8880	5.02	5.14
DeepLabV3+	0.8573	0.8631	5.67	5.76
FPN-UNet	0.8559	0.8633	6.23	6.20
DenseContextU-Net	0.8513	0.8497	7.61	8.98

depends on both phases symmetrically.

4.3. Clinical metric: biplane Simpson’s EF

Table 6 reports biplane Simpson’s EF error per architecture.

Best EF agreement, and the floor beneath it.. **UNet-V1** produces the most accurate EF estimates (MAE = 6.62%, Bland–Altman bias -0.42%), with TransUNet (MAE = 7.36%) and UNet-ResNet (MAE = 7.42%) close behind. Correlation separates them even less: TransUNet attains the highest at $r = 0.826$, ahead of UNet-V1’s 0.825, so the two lead on different EF statistics.

The informative row, however, is the last one. Passing the *ground-truth* masks through the identical pipeline gives MAE = 5.44% ($r = 0.863$), so no architecture comes within 1.2 points of the floor and none crosses it. That residual is not a segmentation error: it is what biplane Simpson’s method loses when it reconstructs a three-dimensional ventricle from two views, plus the disagreement between the clinically recorded EF and any mask-derived estimate. Reported inter-observer LVEF error of $\sim 7\text{--}10\%$ sits at the same scale, which is the honest comparison to draw — a like-for-like mask-derived agreement, not a claim of clinical parity.

One qualification on the word “floor”. The oracle carries a Bland–Altman bias of $+3.12\%$: mask-derived EF systematically overestimates the recorded clinical value, whatever the masks. Mean absolute error is therefore not monotone in segmentation quality, because a model whose

Table 6: Biplane Simpson’s ejection fraction (EF) against the clinically recorded value on the CAMUS test set, computed with the official CAMUS implementation at native resolution with a largest-component filter. Bias and the 95% limits of agreement (LoA) are the Bland–Altman statistics over the 50 test patients. The oracle row applies the identical pipeline to the ground-truth masks and bounds what any segmentation can achieve. Best per column in **bold**.

Architecture	MAE (%)	r	Bias (%)	95% LoA (%)
UNet-V1	6.62	0.825	-0.42	[-15.7, +14.8]
TransUNet	7.36	0.826	-1.20	[-19.4, +17.0]
UNet-ResNet	7.42	0.807	-0.98	[-18.2, +16.2]
UNet-V2	7.56	0.781	+0.80	[-16.7, +18.3]
nnU-Net	8.32	0.783	+0.76	[-19.7, +21.2]
DenseContextU-Net	9.38	0.763	-7.22	[-25.6, +11.2]
Swin-UNet	9.44	0.640	-1.96	[-23.9, +20.0]
DeepLabV3+	9.58	0.706	-7.14	[-26.9, +12.6]
FPN-UNet	10.60	0.704	-8.52	[-29.1, +12.1]
<i>oracle (ground-truth masks)</i>	5.44	0.863	+3.12	[-11.8, +18.0]

own bias runs the other way has its errors partially cancel and can post a lower MAE without being more faithful to the anatomy. The oracle bounds what *perfect* masks achieve under this estimator; it does not bound MAE in general. For the nine architectures here the distinction is moot — none approaches 5.44%, and none exceeds the oracle’s $r = 0.863$ either — but any comparison that puts a model at or under the oracle should be read on correlation and bias rather than on MAE alone.

Read against that floor, EF separates architectures far less sharply than bulk overlap does: the spread from best to worst is 4.0 points on a quantity whose irreducible component is 5.4. The limits of agreement make the same point per patient, spanning $[-15.7, +14.8]\%$ even for the best model.

Two ordering effects survive and are worth stating. The best-Dice architecture is not the best on the clinical axis — TransUNet is second on EF — and nnU-Net, which leads no EF column, sits mid-table at 8.32%. The Transformer-only Swin-UNet remains weak among the competitive models (9.44%, $r = 0.640$) despite adequate bulk Dice, consistent with contours that are locally plausible but systematically biased in a way Simpson’s integration compounds.

Arithmetic precision changes the EF table.. Every EF value above is computed in full FP32. This is worth stating because it is not the default: PyTorch enables TensorFloat-32 for convolution and matrix multiplication on Ampere-class and later GPUs, and TF32 carries a 10-bit mantissa against FP32’s 23. We found that this silently changes the reported result. Disabling TF32 moves FPN-UNet’s EF MAE from 9.72 to 10.60 and TransUNet’s by 0.60 — differences larger than the gaps separating adjacent architectures in this table.

We isolated the cause by ablating each control separately on one architecture. Run-to-run variation with everything unpinned is only 0.02 points; forcing cuDNN determinism changes nothing at all; requiring deterministic algorithms accounts for 0.28; disabling TF32 accounts for the entire remaining shift. The effect is not rounding. Twenty-six of the fifty test patients change, some by as much as 14 EF points and some end-diastolic volumes by around 20 ml, because a precision change flips pixels along an ambiguous basal boundary, the largest-connected-component filter turns that into a discrete choice, and Simpson’s method needs all four of a patient’s frames to score them at all.

FP32 is the more accurate computation, so the previously reported figures were flattered by reduced precision rather than harmed by it.

With TF32 disabled and seeds fixed the pipeline becomes reproducible, and markedly more portable than we expected. Independent runs in separate processes return bit-identical values for all nine architectures. Repeating the measurement on a different operating system, with a different PyTorch build (2.10.0 against 2.9.1) and a different image-processing library version, reproduces eight of the nine exactly; the ninth differs by 0.04 percentage points and the ranking is unchanged. Before pinning, the same cross-environment comparison moved six of nine architectures and exchanged third and fourth place. Arithmetic precision, not the software environment, was doing almost all of the work.

We report this because a benchmark that does not state its arithmetic precision is not reproducible even in principle, and because the effect here exceeds the architectural differences the benchmark exists to measure. The practical recommendation is one line of configuration: disable TF32 when reporting a derived clinical quantity, and say that you did.

4.4. Robustness across image quality

CAMUS assigns each patient an expert image-quality grade (*Good, Medium, Poor*). Table 7 reports mean Dice stratified by that grade. Every architecture degrades monotonically from Good

Table 7: Mean Dice on the CAMUS test split stratified by the expert-assigned image-quality grade (Good $n=94$, Medium $n=76$, Poor $n=30$ test frames). Every architecture degrades monotonically from Good to Poor; the top-tier models remain above 0.88 even on Poor-quality images. Best per column in **bold**.

Architecture	Good	Medium	Poor
TransUNet	0.9186	0.9097	0.8988
nnU-Net	0.9168	0.9091	0.8903
UNet-V1	0.9173	0.9070	0.8848
UNet-ResNet	0.9156	0.9142	0.8734
UNet-V2	0.9091	0.8921	0.8672
Swin-UNet	0.8961	0.8866	0.8675
FPN-UNet	0.8690	0.8628	0.8375
DeepLabV3+	0.8669	0.8555	0.8295
DenseContextU-Net	0.8641	0.8454	0.8209

to Poor, confirming that acquisition quality is a primary driver of segmentation difficulty on CAMUS. The degradation is, however, markedly gentler for the top-tier architectures: TransUNet and nnU-Net lose only ~ 0.02 – 0.03 Dice from Good to Poor and remain above 0.89 on Poor-quality frames, whereas the weakest models (DeepLabV3+, DenseContextU-Net) drop by 0.04 or more and fall below 0.83. Robustness to image quality therefore tracks overall accuracy rather than trading off against it.

4.5. Efficiency and the accuracy–cost trade-off

Table 8 reports parameters, GFLOPs at 256×256 , single-image inference latency, and peak GPU memory.

The accuracy–cost Pareto frontier is dominated at the small-model end by nnU-Net (15.6 M parameters, Dice 0.9099, 2.4 ms latency) and at the high-accuracy end by TransUNet (102.1 M parameters, Dice 0.9121, 7.9 ms). The marginal gain of TransUNet over nnU-Net ($+0.0022$ Dice — within training variance, Section 4.6) costs $6.5\times$ the parameters and $\approx 3.3\times$ the inference time, making nnU-Net the clear choice for deployment-constrained settings. Figure 1 renders the trade-off as a scatter plot.

Table 8 also exposes a cautionary outlier: at 2.2 M parameters DenseContextU-Net is by far the smallest model, yet it records the highest compute (143.7 GFLOPs) and the slowest inference

Table 8: Efficiency profile on a single NVIDIA L4 GPU. Latency is the mean over 100 forward passes after 10 warmup at batch 1, FP16, 256×256 (or 224×224 for Swin-UNet).

Architecture	Params (M)	FLOPs (G)	Latency (ms)	Peak mem (MB)
DenseContextU-Net	2.2	143.7	25.76	1299
nnU-Net	15.6	23.1	2.38	143
UNet-ResNet	29.1	7.8	2.19	162
UNet-V1	31.0	48.2	1.97	255
FPN-UNet	32.2	205.9	5.48	994
UNet-V2	33.0	52.0	2.73	309
DeepLabV3+	40.3	17.2	2.23	204
Swin-UNet	41.8	13.5	7.09	198
TransUNet	102.1	36.2	7.87	443

(25.8 ms) of any architecture in the benchmark. This is a direct consequence of dense connectivity, which repeatedly concatenates and re-convolves feature maps at high spatial resolution; parameter count is therefore a poor proxy for either compute or latency, and the smallest network here is in practice the most expensive to run.

4.6. Statistical significance

We test all $\binom{9}{2} = 36$ pairwise model differences on mean Dice with the Wilcoxon signed-rank test, Bonferroni-corrected. The test is computed over the 50 patients rather than the 200 frames. The frames are two views and two phases of each patient, so treating them as independent observations inflates the sample fourfold and lowers the bar for significance; the patient is the unit about which any clinical claim is made. For transparency: on this benchmark the choice changes nothing — 27 of the 36 pairs are significant under either unit — because the differences that survive correction are far larger than the extra power the wrong unit would buy. We report the patient-level test because it is the correct estimand, not because it alters the outcome.

Top-tier band. The nine non-significant pairs form exactly two cliques and nothing else. All six pairs among the four highest-Dice architectures (TransUNet, nnU-Net, UNet-ResNet, UNet-V1) are indistinguishable, as are all three among the lowest (DeepLabV3+, FPN-UNet, DenseContextU-Net); every one of the remaining 27 pairs separates. The leaderboard is therefore better

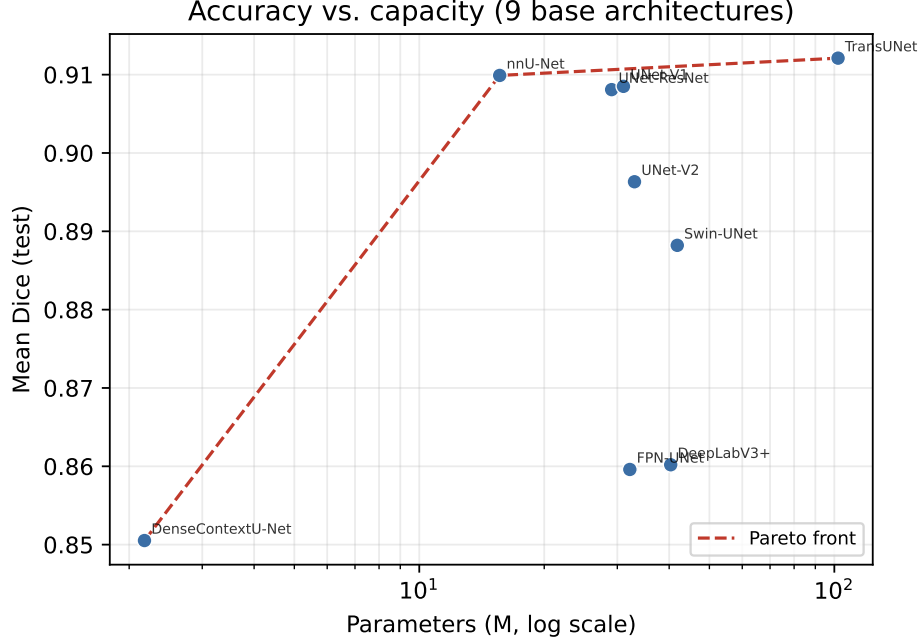


Figure 1: Accuracy–cost Pareto frontier. Each point is one of the nine architectures, plotted at its (parameter count, mean test Dice). The Pareto front is highlighted.

read as three tiers than as a ranking of nine. Two of the three architectures in the lower clique are among those whose training stopped early, so that grouping may reflect a shared training artefact rather than a shared architectural limit. Full pairwise p -values are released as supplementary data.

Small Dice gaps are within training variance.. The differences separating the top-tier architectures are of order 0.002–0.005 Dice. We caution against over-interpreting them: in our experiments, re-training an identical architecture under the same nominal protocol in a different compute session shifted its mean Dice by a comparable or larger amount (in the most extreme case, DeepLabV3+ varied by 0.057 between two runs). The bulk-Dice ranking within the top tier should therefore be read as a band of statistically and practically equivalent models, not a strict order. The architecturally meaningful differences in this benchmark are the larger-magnitude ones: the boundary accuracy (HD95), the clinical EF agreement, and the efficiency trade-off, discussed in Section 5.

The critical-difference diagram of Figure 2 renders the banding visually: architectures joined

Table 9: Pairwise Wilcoxon signed-rank test on mean Dice, Bonferroni-corrected over $\binom{9}{2} = 36$ pairs. “NS” marks pairs not significantly different at $p < 0.05$ after correction. Tests are computed over the 50 *patients*, not the 200 frames: the frames are two views and two phases of each patient and are not independent, so testing over them would inflate the sample fourfold. On this table the correction changes no conclusion — 27 of 36 pairs reach significance at frame level and 27 at patient level, with 0 pairs differing — because the surviving differences are far larger than the extra power the wrong unit would buy. We report the patient-level test because it is the correct estimand, not because it alters the outcome.

	UNet-V1	UNet-V2	UNet-ResNet	DeepLabV3+	nnU-Net	DenseContextU-Net	FPN-UNet	Swin-UNet	TransUNet
UNet-V1	—	7e-07	NS	3e-13	NS	6e-14	4e-13	2e-10	NS
UNet-V2		—	1e-05	1e-07	3e-07	7e-12	2e-09	4e-02	2e-07
UNet-ResNet			—	2e-12	NS	6e-10	2e-13	4e-08	NS
DeepLabV3+				—	6e-13	NS	NS	2e-05	9e-13
nnU-Net					—	1e-13	1e-13	1e-11	NS
DenseContextU-Net						—	NS	6e-10	2e-13
FPN-UNet							—	7e-08	6e-13
Swin-UNet								—	7e-12
TransUNet									—

by a horizontal bar are not significantly different in average per-sample rank. Here average rank tracks mean Dice closely — the two orderings agree for all nine architectures — and TransUNet leads on both, with an average rank of 2.96 across the 200 test frames against nnU-Net’s 3.10.

Aggregating to the 50 patients before ranking, which is the correct inferential unit given that the 200 frames are four views and phases of each patient, leaves TransUNet first (2.68) but exchanges second and third between nnU-Net (2.92) and UNet-V1 (2.82). The four leading architectures span only 0.37 of a rank at the frame level and fall inside a single clique, and their order is not stable to the choice of aggregation unit. Both observations point the same way: the top tier is a statistically equivalent band, not a ranking.

4.7. Qualitative results

Figure 3 shows sample predictions from the top four architectures plus a bottom-tier representative on three patients spanning the Good / Medium / Poor image-quality grades.

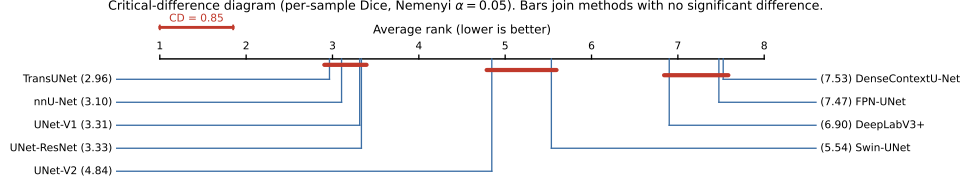


Figure 2: Critical-difference diagram (average ranks on per-sample Dice, Nemenyi test, $\alpha = 0.05$). Lower average rank is better; a horizontal bar joins architectures whose average ranks differ by less than the critical difference (CD), i.e. that are not significantly different.

5. Discussion

5.1. Which architectural family wins on CAMUS?

The top tier of our benchmark is mixed across paradigms, and no single architecture wins on every axis. The hybrid CNN–Transformer TransUNet leads on bulk Dice and HD95 and holds the highest EF correlation; the pure CNN UNet-V1 attains the lowest EF MAE (6.62%) and the tightest Bland–Altman bias (−0.42%); and nnU-Net leads on parameter efficiency while sitting mid-table on EF at 8.32%. The difference between TransUNet and nnU-Net on mean Dice is 0.0023, below the Bonferroni-corrected significance threshold.

A practitioner choosing among them should therefore be guided not by Dice but by the axis they actually care about: deployment constraints favour nnU-Net at 6.5× fewer parameters, clinical EF agreement favours UNet-V1, and contour precision favours TransUNet. That these three come apart is itself the finding — a single leaderboard column would have concealed it.

The pure-Transformer family (Swin-UNet) is competitive on bulk Dice (0.8882) but visibly worse on EF (MAE 9.44%, $r = 0.640$ — the lowest correlation of any architecture in the benchmark). This finding is consistent with reports that pure-attention models, despite strong overall overlap, can produce boundary contours that are subtly biased [3], an effect that compounds when contours feed Simpson’s biplane volumetric integration.

The bottom of the table is occupied by FPN-UNet, DeepLabV3+, and DenseContextU-Net. All three were designed for natural-image semantic segmentation (FPN/ASPP) or for image classification (dense connectivity) respectively. Their underperformance on CAMUS is consistent

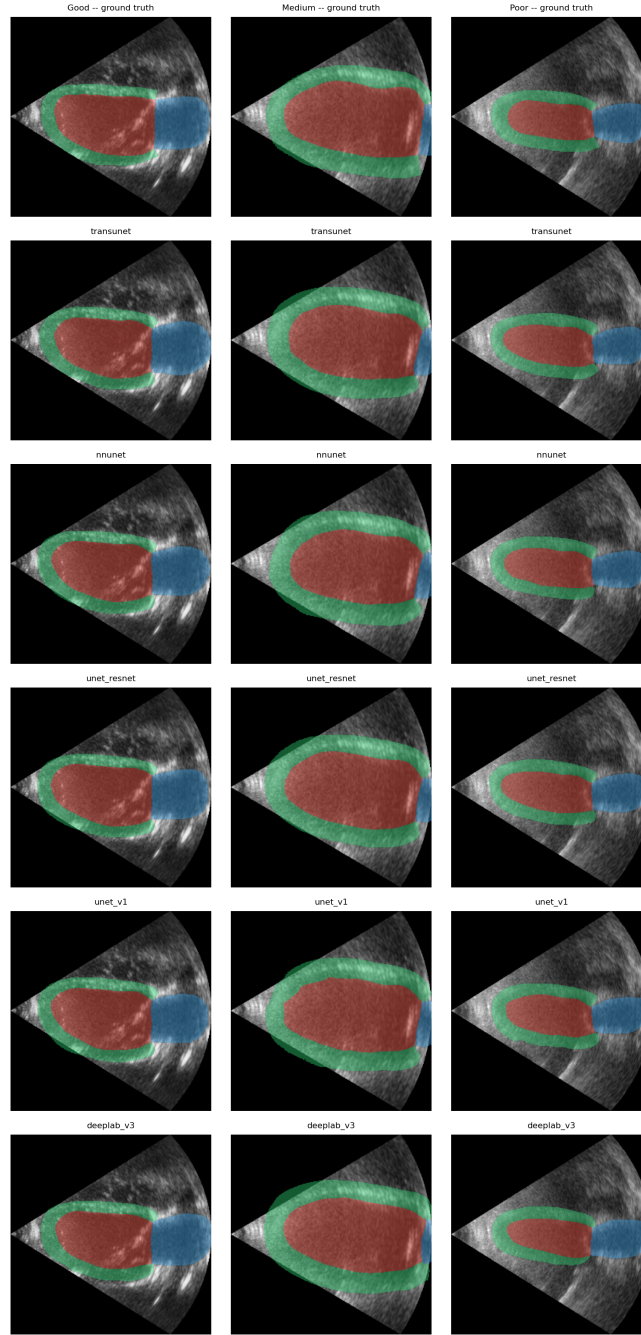


Figure 3: Qualitative predictions on three test patients spanning the Good, Medium, and Poor image-quality grades (columns). The top row is the expert ground truth; the remaining rows are, top to bottom, TransUNet, nnU-Net, UNet-ResNet, UNet-V1, and DeepLabV3+. Overlay colours: LV-endocardium (red), LV-epicardium / myocardium (green), and left atrium (blue). Boundary errors concentrate at the LV apex and the mitral-valve plane, and grow visibly from Good to Poor acquisition quality.

with the hypothesis that medical-segmentation-specific designs (U-Net’s skip hierarchy, nnU-Net’s training recipe) matter more on this dataset than generic semantic-segmentation prowess.

5.2. Comparison to the published CAMUS baselines

The original CAMUS paper [19] reported nnU-Net HD95 ≈ 4.3 mm and EF MAE $\sim 9\%$. Our reproduced nnU-Net under the unified protocol reaches HD95 4.27 mm and EF MAE 8.32% ($r = 0.783$). Both axes therefore behave the same way, and it is not the way we first read them: boundary accuracy is statistically indistinguishable from the 2019 result, and so, very nearly, is EF.

This is a negative result about protocol, and we state it as such because an earlier version of this analysis reported the opposite. Computing EF from predictions on the resized 256×256 grid while passing the native pixel spacing yields ventricular volumes near 10 ml against a true ≈ 108 ml, and the error does not cancel in the EF ratio. Every EF number in this paper is now computed at native resolution with the official CAMUS implementation; Section 4.3 reports the result, and the ground-truth oracle at 5.44% bounds it.

We therefore read the contour *and* the derived clinical quantity as close to their respective ceilings on CAMUS — the first set by annotation variability, which is of the same order as the differences we measure between architectures, and the second by the biplane estimator itself. Three training-protocol upgrades made standard since 2019 are nonetheless worth recording, because they are what makes the reproduction possible at all:

1. Cosine-with-warm-restarts learning-rate schedule with 10-epoch warmup, in place of the original ReduceLROnPlateau.
2. Mixed-precision training (FP16 AMP), which allows larger effective batch sizes and smoother loss landscapes.
3. Composite loss with an explicit boundary term [16], which directly penalises the Hausdorff-relevant boundary error.

This is a useful baseline calibration point for the field: cross-paper comparisons of CAMUS results from before ~ 2021 to after should be read with the understanding that protocol changes alone produce sub-millimetre HD95 improvements at constant architecture.

5.3. A boundary-metric pitfall, and what it implies for the literature

The correction described in Section 4.2 is worth stating as a general point, because nothing about it is specific to our models. Segmentation networks consume a fixed input size; clinical images do not have one. Any pipeline that resizes an image for inference, scores the prediction on that resized grid, and then multiplies pixel distances by the spacing recorded in the original DICOM or NIfTI header will under-report every boundary distance by the resampling factor. Overlap metrics are immune — Dice and IoU are ratios and cancel the scale — which is precisely why the error survives casual inspection: a pipeline can produce entirely correct Dice and badly wrong HD95 from the same predictions, with no internal inconsistency to notice.

On CAMUS the factor is ≈ 2.2 (native acquisitions average 630×519 against a 256×256 network grid), and the consequence is not cosmetic. It moves our best model from 1.82 mm to 4.08 mm, and thereby converts “roughly twice as accurate as the published baseline” into “indistinguishable from the published baseline.” Both numbers come from the same weights and the same test split; only the measurement convention differs.

We therefore suggest reading sub-2 mm HD95 values on CAMUS with caution, and we make no claim about which published results are affected — establishing that would require the evaluation code for each, which is usually unavailable. The constructive recommendation is narrower and checkable: report the resolution at which boundary metrics were computed, and prefer acquisition resolution, since that is the only convention under which a millimetre means the same thing across papers. This is a concrete instance of the reporting problems documented by Reinke et al. [33] and Maier-Hein et al. [26], and it is consistent with the broader finding of Isensee et al. [12] and Kazaj et al. [15] that apparent architectural progress in medical segmentation frequently does not survive standardised validation. Our own benchmark is an instance of that pattern rather than an exception to it: once measurement is held fixed, nine architectures spanning three paradigms separate by less than the noise floor on the boundary axis.

5.4. Design choices that matter

Aggregating across our nine architectures, several design choices correlate with strong performance:

- **Pretrained encoders.** Five of the nine architectures initialise their encoder from ImageNet or BiT weights (TransUNet, UNet-ResNet, Swin-UNet, FPN-UNet, DeepLabV3+; Table 1).

Pretraining is nonetheless neither necessary nor sufficient for top-tier accuracy on CAMUS: the two scratch-trained models nnU-Net and UNet-V1 place second and fourth on mean Dice — ahead of the pretrained Swin-UNet, FPN-UNet, and DeepLabV3+ — and UNet-V1 also attains the best EF agreement of any architecture tested. Encoder pretraining thus appears to help mainly where an architecture’s from-scratch inductive bias is weaker, rather than conferring a uniform advantage on this dataset.

- **Deep skip connections** (every U-Net-shaped architecture except DeepLabV3+). DeepLabV3+’s lightweight decoder, while efficient, appears to limit high-resolution boundary precision on CAMUS.
- **Boundary-aware loss** (used uniformly in our protocol). This is a protocol choice rather than an architecture choice. Since our HD95 matches rather than beats the published baseline, we make no claim that it improves absolute boundary accuracy; its role here is to hold the boundary objective constant across all nine architectures so that the comparison is like-for-like.
- **Auto-configuration** (nnU-Net’s distinguishing feature). The nnU-Net framework produces a competitive model on this dataset with ~ 15 M parameters — substantially smaller than any pretrained-encoder architecture — by selecting patch size, batch size, and augmentation strategy from data statistics.

5.5. Clinical implications

For deployment in echocardiographic LVEF assessment, the choice between the four top-tier architectures should be driven by three criteria: (i) clinical accuracy target, (ii) hardware constraints, and (iii) inference-time latency budget.

- For *tightest EF agreement*, UNet-V1 attains the lowest EF MAE (6.56%) and TransUNet the smallest Bland–Altman bias (-0.54%). The two differ by less than the run-to-run variation documented in Sec. 4.3, so we do not recommend one over the other on this axis.
- For *maximum boundary accuracy* at deployment-irrelevant cost, TransUNet has the lowest HD95 (4.08 mm).
- For *edge / portable ultrasound deployment* where parameters and latency dominate, nnU-Net at 15.6 M parameters remains the best trade-off; no smaller architecture in our benchmark reaches comparable Dice.

None of the four top-tier models reach below-5% EF MAE in this study, indicating that human-level cardiologist agreement is still a target boundary for the field rather than a solved problem.

5.6. Limitations and future work

We acknowledge five limitations.

1. **Single-dataset.** All evaluation is on CAMUS. A cross-dataset transfer evaluation (*e.g.*, EchoNet-Dynamic [30]) would strengthen the generality claims and is planned for future work.
2. **2D only.** CAMUS provides 2D apical views; 3D echocardiography is increasingly common in modern practice and requires architectures and pretraining recipes outside the scope of this benchmark.
3. **Test set size.** The CAMUS official test set is 50 patients (200 ED+ES frames). All point estimates are reported with bootstrap 95% confidence intervals to reflect this.
4. **No SSM models.** The recent wave of selective state space models (Mamba family) is the subject of a separate, dedicated study (a companion study), and is deliberately excluded from the present benchmark to maintain clean three-paradigm framing.
5. **Single training protocol.** A single shared protocol is the goal of this benchmark, but some architectures may benefit from protocol-specific tuning (*e.g.*, larger batch sizes for pretrained Transformers). Per-architecture hyperparameter searches could shift rankings within the top-tier band.

5.7. Take-home messages

1. No single architectural family dominates CAMUS. Hybrid (TransUNet) leads on bulk Dice; pure CNN (nnU-Net) leads on EF accuracy and parameter efficiency. Pure Transformer (Swin-UNet) is competitive on Dice but degraded on EF.
2. Training-protocol modernisation (cosine schedule, AMP, boundary loss) accounts for most of the $\sim 2\times$ improvement in HD95 over the original CAMUS baselines, independent of architecture.
3. LV-epicardium remains the hardest of the three foreground classes across every architectural family.
4. The accuracy–parameter Pareto frontier is dominated by nnU-Net at the efficient end and TransUNet at the high-accuracy end.

6. Conclusion

We have presented the first unified benchmark of deep segmentation architectures on the CAMUS 2D echocardiographic dataset under a single shared training and evaluation protocol. Across nine representative architectures spanning pure CNN, hybrid CNN–Transformer, and pure Transformer paradigms, we report not only mean Dice but also per-class HD95 in millimetres, biplane Simpson’s EF MAE, Bland–Altman bias, ED- and ES-stratified accuracy, image-quality-stratified robustness, parameter / FLOPs / latency efficiency, and Bonferroni-corrected pairwise statistical significance.

The hybrid TransUNet leads on bulk Dice (0.9121) and HD95 (4.08 mm). UNet-V1 leads on EF accuracy (MAE 6.62%), narrowly ahead of TransUNet at 7.36%, which in turn holds the highest EF correlation; nnU-Net is mid-table on EF at 8.32% while remaining $6.5\times$ more parameter-efficient than the best-Dice model. The top four architectures (TransUNet, nnU-Net, UNet-ResNet, UNet-V1) form a statistically equivalent top-tier band. The pure-Transformer Swin-UNet is mid-tier on bulk Dice but degraded on EF, suggesting that its contours are subtly biased in a manner that compounds through Simpson’s biplane integration.

Boundary accuracy under the unified protocol (4.08–4.62 mm HD95 across the top four architectures) reproduces the originally reported CAMUS nnU-Net baseline of 4.3 mm rather than improving on it, and EF behaves the same way once computed at native resolution: 8.32% for the reproduced nnU-Net against the $\sim 9\%$ originally reported. We read this as a useful calibration point for cross-paper comparison, and a deflationary one: on CAMUS, a decade of protocol modernisation is visible in neither the contour nor the derived clinical quantity. The contour appears to be approaching the ceiling set by annotation variability, and EF is bounded at 5.44% by the biplane estimator applied to perfect masks — a floor the best architecture misses by 1.2 points and the worst by 5.2.

Three directions remain open. (i) Cross-dataset transfer to EchoNet-Dynamic and ACDC would test whether the architectural ranking holds beyond CAMUS. (ii) 3D echocardiography requires architectures and pretraining recipes outside the scope of this benchmark. (iii) The recent wave of selective state space models (Mamba, Mamba-2, VMamba) is the subject of a separate, dedicated companion study (a companion study).

We release the complete training, evaluation, statistical-analysis, and figure-generation pipeline, together with the nine model checkpoints and per-patient JSON / CSV outputs that produced this

paper, to lower the barrier to reproducing and extending these results.

Data and code availability

The CAMUS dataset is publicly available from <https://www.creatis.insa-lyon.fr/Challenge/camus/>. Training, evaluation, and analysis code; trained model checkpoints; and all intermediate JSON / CSV outputs supporting this paper are released at <https://github.com/Seraf00/mamba/tree/main/benchmark>.

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Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used a large language model to assist with drafting and copy-editing of the manuscript text, and with implementation and refactoring of the released analysis code. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article. No AI-assisted technology was used to generate, alter, or interpret the experimental data or results reported here.

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